Interpretable Machine Learning: A brief survey from the predictive maintenance perspective

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***Abstract*—In the field of predictive maintenance (PdM), machine learning (ML) has gained importance over the last years. Accompanying this development, an increasing number of papers use non-interpretable ML to address PdM problems. While ML has achieved unprecedented performance in recent years, the lack of model explainability or interpretability may manifest itself in a lack of trust. The interpretability of ML models is researched under the terms explainable AI (XAI) and interpretable ML. In this paper, we review publications addressing PdM problems which are motivated by model interpretability. This comprises intrinsically interpretable models and post-hoc explanations. We identify challenges of interpretable ML for PdM, including (1) evaluation of interpretability, (2) the observation that explanation methods explaining black box models may show black box behavior themselves, (3) non-consistent use of terminology, (4) a lack of research for time series data, (5) coverage of explanations, and finally (6) the inclusion of domain knowledge.**

***Index Terms*—machine learning, XAI, interpretable ML, explainability, transparency, trust, introspection, explanation-awareness, predictive maintenance, RUL, time series**

I. Introduction

The term explainable AI (XAI) was first mentioned by [1] to describe a system with the ability to explain the behavior of simulation game entities controlled by AI [2]. However, the first instance of explainable AI systems dates back to the mid of the 1970s, where explanations of expert systems were studied [3]. Since the rise of deep learning (DL) starting from 2012 onward, explainability and interpretability of machine learning models has increasingly developed as a research field.

A popular definition by DARPA states the goal of XAI as "to create a suite of new or modified ML techniques that produce explainable models that, when combined with effective explanation techniques, enable end users to understand, appropriately trust, and effectively manage the emerging generation of AI systems. The target of XAI is an end user who depends on decisions or recommendations produced by an AI system, or actions taken by it" [4]. Another definition by [5] states that XAI aims "to ensure that algorithmic decisions as well as any data driving those decisions can be explained to end-users and other stakeholders in non-technical terms."

The lack of explainability or interpretability often manifests itself in the absence of trust in the model in industrial settings [6]–[8]. The lack of successful applications of data-driven models in practice is often attributed to the absence of trust [9]. In other words, trust in models can be deemed as a prerequisite for the successful application of data-driven models [10], [11].

In the field of predictive maintenance (PdM) the use of machine learning has gained importance over the last years as shown e.g. in [12]–[19]. While preventive maintenance relies on pre-scheduled maintenance intervals like fixed time or usage intervals, predictive maintenance (PdM) aims to avoid premature and costly repairs, while at the same ensuring a timely repair prior to a failure [14], [20], [21]. Advanced methods estimate the remaining useful life (RUL), see e.g. [22]. In addition to the use of physical models, data-driven methods are used – in some cases hybrid models are formed of physical and data-driven models. Accompanying this development, an increasing number of non-interpretable methods have been proposed to address PdM problems, see e.g. [12].

Even though the problems of non-interpretable methods were first noted shortly after the first proposals of DL in PdM [23], recent reviews of data-driven PdM [12], [14], [24]–[30] suggest that this problem is now even more prevalent, due do the extensive use of DL models in PdM. However, to the best of the authors' knowledge, no review exists which explicitly surveys the state-of-the-art and status of interpretable ML in data-driven PdM; typically, the lack of interpretability is rather treated as a consequence of the data-driven methods, e.g. DL.

Hence, we see a need to review work on explainable AI and interpretable machine learning for PdM. Consequently, in this paper we review papers addressing PdM problems, motivated by model interpretability. This comprises approaches with intrinsically interpretable models and post-hoc explanations.

In this paper, we make the following contributions:

1) We relate the different definitions and categorizations in the field of XAI and interpretable ML.
2) We identify interpretable machine learning as a research field of increasing importance for PdM. Hence, we review and discuss current work in the field.
3) We identify challenges of interpretable ML for PdM.

II. Explainable Artificial Intelligence

In addition to methodological and technical challenges, the lack of formal and widely-accepted definitions of terms surrounding the concepts of XAI is one of the most criticized issues of the research field [31], [32].

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To distinguish between explainability and interpretability, the definition of [33] can be used: "systems are interpretable if their operations can be understood by a human, either through introspection or through a produced explanation." This leads to the notion that interpretability is a passive characteristic, that describes whether a model and the decisions derived through it, make sense to a human – also defined as the transparency of a model [9]: Here, a distinction is made between models that provide interpretability themselves via introspection and models which require an explanation. This is a vital distinction that is also made in many other publications such as in [34], [35]. Hence, explainability is viewed as an active characteristic of a model or method generating an explanation [9]. It is important to note, that interpretability therefore comprises explainability [36], [37].

III. Intrinsically interpretable Models

Every interpretable model [33] is also explainable, per default through introspection or through additional efforts taken to produce an explanation. However, the definition in [33] can be refined to clarify when a model can be considered as interpretable. To do so one can use the definition of transparency from [38], introducing three levels of transparency:

- **Simulatability** is considered as the strictest sense in which a model can be transparent [39]. If a person that is given the input data and the parameters of a model can produce a prediction by going over every calculation of the model in reasonable time, a model is deemed transparent at the simulatability level [40]. This favors sparse models [9].
- At the **decomposability** level, each part of the model (input data, parameters, calculations) must be interpretable [41], [42]. This requires the features to be interpretable, rendering models trained on uninterpretable features non-transparent and not interpretable [38].
- **Algorithmic transparency** refers to how model parameters or prediction are generated [43]. For a linear model, e.g., the shape of the loss surface is understandable and can be shown to converge to a unique solution. This is not the case e.g., for neural networks in which heuristic optimization algorithms are used.

In this paper, we consider a model as interpretable by introspection if it satisfies the definition of transparency at the decomposability level. While we consider the raw input data to be transparent, often feature engineering methods are used. This might change the level of transparency of the data itself [31], [34]. The primary motivation for choosing the decomposability level over the algorithmic level, lies in the popularity of deep models for feature learning. We argue that using algorithmic transparency opens up the possibility of categorizing combinations of non-interpretable models with interpretable ones as transparent and therefore interpretable. As interpretability distinguishes between models that are interpretable per introspection and models that need additional explanations, the terms **intrinsic interpretability** and **post-hoc explanations** will be used from now on [31], [44].

IV. Post-hoc Explanations

Post-hoc explanations provide interpretability after a model has been constructed [35], [45], [46]. There is a variety of post-hoc explanation methods, along different dimensions: **global methods** explaining the entire model and **local methods** providing explanations for single predictions [2], [47]. Further, post-hoc explanation can be either **model-agnostic** or **model-specific** [2]. Model-specific methods are restricted to a specific model or a class of models [31]. Model-agnostic methods on the other hand can be applied to any model, as long as the features and model outputs match the chosen explanation method [9]. Model-agnostic methods rely on analyzing pairs of feature inputs and outputs; as per definition they do not have access to specific model parameters such as weights or structural information [31], [37]. Model-specific methods may access model parameters and structural information. They thereby trade-off a more narrow applicability in favor of more insights into the model [44]. Note that intrinsically interpretable models are by definition model-specific [2].

An alternative categorization can be the type of explanation: In [9] explanations are subdivided into textual, visual, example-based, feature relevance-based explanations as well as explanations by simplification. In [41] numerical, rules, textual, visual and mixed explanations are distinguished, while [2] categorizes explanations into visualization, knowledge-extraction, influence methods and example-based. In our review, no publication used textual explanations. Hence, ignoring textual explanations, the categorizations of [2] and [9] are almost identical: visual and example based explanations do coincide between the two publications. Furthermore, the terms of feature-relevance and influence methods both refer to the same concepts. Additionally, knowledge-extraction explanations by [2] encompasses the explanations by simplification definition given by [9]. Thus, we categorize explanation types into:

- **Visual explanations** provide explanations via graphical plots and summaries, e.g. heatmap-overlays of the input data, decision boundary plots, visualizations of model parameters or feature importances [2].
- **Example-based explanations** select specific data instances as key examples to explain the model behavior [9]. Representative examples, called prototypes, or underrepresented examples, called criticism, can be selected [47]. Another example-based explanation are counterfactual examples, describing the changes to a sample that would be necessary to alter its prediction [48].
- **Feature-relevance explanations** quantify the features' contributions [9]. They are often combined with visual explanations to improve comprehensibility.
- **Knowledge-extraction explanations** transfer the hidden knowledge in the model to a white-box model or extract the hidden knowledge into more comprehensible and transparent knowledge through e.g. symbolic representations and rules [2], [47], [49]. A further knowledge-extraction approach are surrogate-models which approximate the predictions of the model [31].

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V. Interpretable ML in PdM: a review

A challenge in surveying interpretable ML/XAI is the lack of consistent terminology. Therefore, the search was dynamically extended by synonyms to the terms explainable and interpretable, cf. [41], [50], also including specific XAI methods. Finally, a forward and backward search was conducted for the identified publications. We searched IEEExplore, ScienceDirect, ACM, Springer, MDPI, and Scopus. In total, this resulted in 40 papers. Out of the 40 identified publications 20 used models which are considered transparent and hence intrinsically interpretable. Table I shows the classification of the identified publications according to the criteria identified in the two previous sections.

The application domains are very diversified and range from milling tools [51] and gearboxes [52] over to heat recyclers [53], hard disks [54] and semiconductor manufacturing [55]. All approaches use time series data. The majority of the data sets used are not publicly available. Hence, the performance of the approaches are not quantitatively compared.

Papers by PdM tasks: The PdM task **diagnosis** is addressed by 22 of the papers, while 9 and 8 do exclusively address **prognosis** and **detection**, respectively. One publication covers diagnosis as well as prognosis. Interestingly, the prognosis task is much more favored by intrinsically interpretable approaches as opposed to post-hoc explanations. Only three cases showcase the application of post-hoc methods to the prognosis task, whereas six intrinsically interpretable approaches were used for prognosis. For the task of diagnosis it is more evenly distributed, however, the authors seem to favor the applicability of post-hoc methods for diagnosis over intrinsically interpretable approaches as 14 out of the 23 publications employ post-hoc explanations. The detection task is more evenly split between both methods.

A. Intrinsically interpretable models

The intrinsically interpretable approaches do, however, not show a preference for a single model. The surveyed methods build on knowledge-bases [56] and ontologies [55], to simple decision trees [57] over to Wiener processes [58] and Particle filters [59]. A difference can be observed between the models used for diagnosis and detection versus the models used for prognosis. The models for diagnosis and detection are based on knowledge, rules and decision trees, while the intrinsic models for prognosis consist of mostly stochastic models, such as the Wiener process and hidden Markov models. In the case of diagnosis, four of the identified approaches do make use of knowledge-based methods, in the form of knowledge-bases or ontologies, and various ML methods to extract knowledge from the data and use it in combination with the knowledge-base. The resulting inference is done via rules and considered transparent, hence interpretable. However, these approaches entail a much larger fraction of expert labor than other approaches, as the initial knowledge-bases are built by hand. For prognosis, no specific approach was used more than once. However, the prevalence of stochastic models, such as the Wiener process and hidden Markov models can be observed.

B. Post hoc-explanations

The post-hoc methods are almost evenly split between model-specific explanation methods, in this case primarily Layer-wise Relevance Propagation (LRP) [60] and class activation mapping-based (CAM) [61], [62] approaches, and model-agnostic explanation methods, namely local interpretable model-agnostic explanations (LIME) [63] and SHapley Additive exPlanations (SHAP) [64]. Post-hoc methods were most often applied to convolutional neural networks (CNNs), which were also the most popular model encountered in the publications regardless of the explanation. CAM-based methods were only applied to CNNs, as they are model-specific to them. LRP was applied to CNNs by four out of the five publications. LIME was however the only post-hoc method that was not applied to CNNs at all. Furthermore, LIME and SHAP showed no concentration on a particular model and were applied to Long short-term memory networks (LSTMs), Autoencoders, multilayer perceptrons (MLPs), Support-vector machine (SVMs), Random Forests and Gradient Boosted Trees.

Univariate diagnosis: In [83] a 1D-CNN is used to classify faults in linear motion guides based on vibration signals. They use a gradient-weighted class activation mapping (Grad-CAM) explanation in the frequency domain arguing that different errors manifest themselves via different frequency patterns that are not observable to humans in the time domain. However, they aimed to keep the simplicity of the time domain vibration signal as the input for the 1D-CNN. A similar approach was used in [84] with a 1D-CNN to diagnose bearing faults based on their vibration signal. Different from [83] they transform the vibration signal via a fast Fourier transformation (FFT) before using it as input for the CNN. Also, the Grad-CAM explanation method is used. Both input data the explanations are therefore provided in the frequency domain. The visual explanation differs slightly between both approaches. In [85] a 1D-CNN is utilized to diagnose faults of an electric motor based on the motor's magnetic field. They use a CAM explanation which is visualized via a heatmap overlay in the time-domain.

Two further CAM based methods are proposed by [51] and [88] who analyze the condition of milling tools and gas turbines respectively. Both approaches transform the univariate signal into a two dimensional structure. In [51] a recurrence plot is used, while in [88] features are extracted and fused into a 2D structure. 2D-CNNs are then used as classifiers. The resulting predictions are explained via the Grad-CAM method and visualized via a heatmap over the two dimensional structures. Besides CAM explanation, which are only applicable to classification, LRP was also only used for diagnosis, i.e. classification. Interestingly, apart from [82] who used LRP to explain the predictions of a LSTM, LRP was only used on CNNs and in a similar way to CAM explanations. Taking a closer look at the application of CAM and LRP explanation, they are almost used interchangeably, in the sense that they are used for the same kind of explanations. In [81] a wavelet

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TABLE I

OVERVIEW OF REVIEWED PAPERS. THE PUBLICATIONS MARKED WITH ([1, 2]) ARE CATEGORIZED INTO MULTIPLE SUBCATEGORIES. (FR=FEATURE RELEVANCE, V=VISUAL, KE=KNOWLEDGE-EXTRACTION; DATA: UVTS=UNIVARIATE TIME SERIES, MVTS=MULTIVARIATE TIME SERIES)

Paper	PdM task	Data	Agnostic/specific	Explanation method	Explanation Type
Intrinsically interpretable models:					
[65]	detection	uvts	intrinsic	Empirical Mode Decomposition, L-Kurtosis	Introspection
[66][1]				Decision Tree	
[6]				Regularized Logistic Regression	
[67]				CUMSUM, Gaussian Graphical Model	
[68]	diagnosis	uvts		Decision Tree, k-NN	
[69]				Autoregressive Model (GNAR-GARCH), k-NN	
[70]				Logical Analysis of Data (Rules + Binary Database)	
[55]		mvts		Ontology	
[56]				Ontology	
[71]				Ontology	
[57]				Decision Tree	
[72]				Granger Causality, Transfer Entropy	
[73]				Matrix Profile, Ontology	
[74]	prognosis	mvts		Quantitative Association Rule Mining	
[75]				Subspace-vector Autoregressive Model	
[59]				Particle Filter	
[58]				Wiener Process	
[76]				Mixed Effect Model	
[77]				Hidden Markov Model	
Post-hoc explanations:					
[78]	detection	mvts	specific	Feature Importance	FR + V
[79]			agnostic		FR
[66][1]				LIME	FR + V
[53]					
[80]					
[52]	diagnosis	uvts	specific	LRP	
[81]					
[82]					
[83]				Grad-CAM	V
[84]					
[51]				CAM-based	
[85]					
[86]				CNN with visualization of wavelet convolutional layer	
[87]				Rule Extraction, Feature Importance	KE, FR
[88]		mvts		CAM-based	V
[89]				LRP	FR + V
[90]					
[91]			agnostic	SHAP	
[54]					
[92][2]					
[92]	prognosis	mvts			
[93]					
[94]				LIME, Feature Importance	

transform is applied to the vibration data of induction motors followed by a 2D-CNN. The predictions are explained via an overlay of the two-dimensional data, similar to [51]. A similar approach was used for vibration data of gearboxes [52]. Both approaches follow the same methodology, but differ in network configuration and visualization. The authors of [82] use a LSTM to classify rolling-element bearings. LRP is used with a heatmap visualization in the time-domain. A similar approach with a different explanation method is presented by [91] who use the Kernel-SHAP method to explain predictions of a 2D-CNN on univariate vibration signals.

Multivariate detection & diagnosis: For the use on multivariate data, SHAP and LIME are the predominantly used post-hoc explanation methods. Furthermore, LRP as well as some unnamed feature importance methods were also used. SHAP was the most diversely used method and was applied to a LSTM, CNN, a combination of CNN and LSTM, and gradient boosted trees. SHAP was used by [54] to locally explain the importance of features for the diagnosis of hard-drive failures, based on 9 variables. In [92] SHAP is used to explain the global effect of eight different variables on the thickness of steel beams, distinguishing between three classes. Other local feature importance explanations are proposed by [78] and [79]. The explanations by [79] and [87] are the two feature relevance explanations encountered, which were not visualized. In [66] a random forest model for railway switch diagnosis is explained in two ways: globally via the drop-feature importance score, which requires retraining the model several times, and locally via LIME. The detection of engine faults is addressed in [80] based on statistical features extracted from 123 sensors, using a one-class SVM with LIME. LIME is also used in [53] to explain the predictions

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made by an MLP and an SVM for the fault detection in heat recyclers. The explanations from LIME were visualized with bar plots depicting feature importances. In these cases, LIME and SHAP were used very similarly, i.e. SHAP is used mostly for local explanations, and tend to produce the same kind of explanation, however, visualized differently. The authors of [89] provide a different kind of feature importance explanation using LRP to explain the fault detection with a 2D-CNN that was trained on 27 sensor signals of a proton synchrotron booster inside a particle accelerator. The resulting explanation is delivered via 2D monotone heatmap, which intuitively resembles the two dimensional Grad-CAM and LRP explanations of univariate data. A explanation for multivariate data which differs from the previously given explanations is proposed by [90], who use LRP to diagnose faults in induction motors, based on three sensors. The input data of two acceleration sensors and a current sensor is transformed via a FFT and fed into a 2D-CNN. The resulting LRP explanation are given as heatmap visualizations of the frequency domain input for all three sensors, i.e. three plots are generated for the explanation of a single prediction.

Multivariate prognosis: There appears to be a lack of post-hoc explanations for the task of prognosis. For the explanation of prognosis results, SHAP was used twice, whereas LIME was used once. The post-hoc methods were solely used for multivariate data. A stacked architecture of a 1D-CNN, a LSTM and a bidirectional LSTM was used in [93] to perform RUL prognosis on turbofan engines. They showcase the use of explanations via SHAP force plots and SHAP decision plots of all the available input signals. The other approach [92] uses SHAP for a diagnosis and a prognosis task in steel production. The goal was to predict the thickness of steal beams manufactured, in order to perform maintenance when the beams deviate from the specifications. They use gradient boosted trees as a model and a multivariate time series of process parameters as input. TreeSHAP was used to explain the predictions. Furthermore, [94] analyze the RUL of bushings based on multiple sensor signals with various models, from which gradient boosted trees emerged as the best. They fitted linear regressions on the predictions using LIME to obtain local explanations and permutation importance to calculate global feature importance.

VI. Discussion and Open Challenges

The following overriding conclusions can be drawn from the reviewed papers: First, the number of identified PdM papers addressing interpretable ML still makes up a small proportion of PdM papers. The importance of the topic is still to be recognized in this field. Second, over all PdM tasks, i.e. detection, diagnosis and prognosis, combined intrinsically interpretable models and post-hoc explanations, are equally distributed among the surveyed papers. While many post-hoc approaches were applied to diagnosis and detection tasks, the vast majority of papers on prognosis chose intrinsically interpretable models. Third, explanations for the use on univariate time series data are without exception all local, i.e. used to explain a single prediction. Fourth, the explanations are displayed visually, usually on the input data, with one exception to that. Four types of visual explanations were used: visualization in the time-domain, in the frequency domain, in the time-frequency domain, or in a generated vibration image. Contrary to that, models using multivariate data are either explained through feature importance by treating each univariate time series as a feature or, in some cases, by quantifying the importance of the entire multivariate time series. Fifth, the encountered post-hoc explanations are either visual explanations or feature relevance explanations that are visualized – no textual explanations nor knowledge-extraction explanations were identified in PdM-related publications. Figure 1 shows the overall categorization of the reviewed papers.

From the surveyed papers and the discussion, we deduced the following challenges in order to inspire future research:

Challenge 1: Evaluation of model interpretability: While for ML models there are acknowledged performance measures, the evaluation of model interpretability is currently a challenge. This has been postulated in [47], but currently no widely accepted quantitative for this purpose has been proposed. The reviewed papers evaluate their interpretability in a subjective manner, most of them on the basis of one or several examples of local explanations or simply depict the global feature importance. This might be unintentionally biased by the experience and the logical reasoning of the examiner. User studies, perhaps the most reliable performance measure, were not conducted in the reviewed papers.

Challenge 2: Some methods to explain black box models might be black box themselves: Explaining complex ML models calls for sophisticated ways to explain them. Explanation methods come with assumptions, e.g. local linearity or feature independence, and with hyperparameters, e.g. to control approximations of computationally highly expensive explanations. This increases the chance of applying the explanation models in a way that generates invalid explanations.

Challenge 3: Non-consistent use of terminology: As we experienced during the search for research papers, the terminology in this young research field is not used in a consistent way. A study conducted by [50] confirms that this is not only a problem in research, but also in industrial and business settings. She interviewed 60 stakeholders in XAI (machine learning researchers, data analysts and decision maker, i.e. end users). As a primary finding, a lack of standardized terminology was noted. Furthermore, it was noted that many professionals use different terms as synonyms or the same term as antonyms [50]. The reviews [2], [47] have proposed a set of terms and definitions that might help to unify the use of the terminology.

Challenge 4: Lack of research for interpretable ML for time series: The vast majority of interpretable ML research is conducted for images. Yet, in PdM time series are frequently encountered. In fact, all of the reviewed papers used univariate or multivariate time series. We argue, that interpretable ML for time series is yet an open research direction. Research in that direction has been conducted e.g. in [95].

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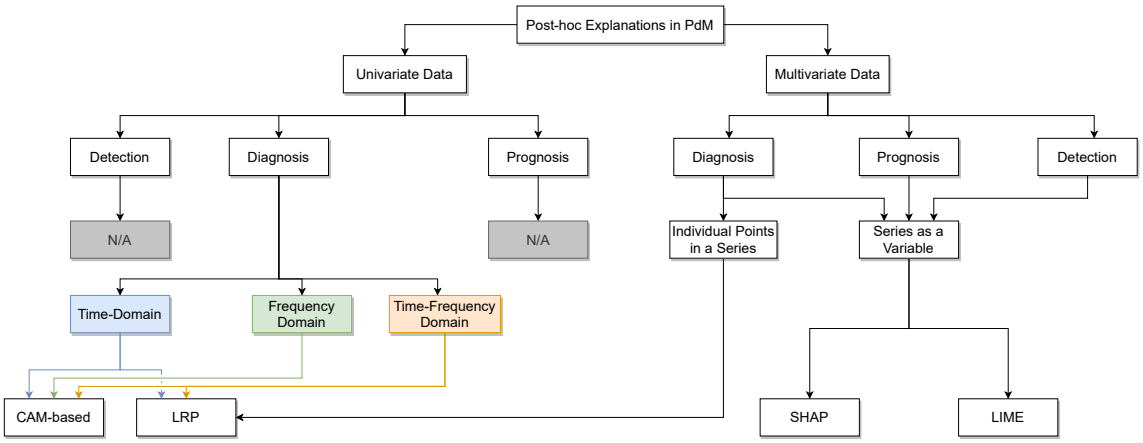


Fig. 1. Categorization of post-hoc explanations used in the surveyed PdM papers.

Challenge 5: Coverage of explanations w.r.t. black box models: Model-agnostic approaches for explaining black box models can typically generate explanations for specific data spaces and distributions. However, it is currently not clear, how a comprehensive coverage of the relevant situations can be obtained, e.g., via simulation or testing/assessment procedures, in particular in practical application settings.

Challenge 6: Integration of domain knowledge into XAI: While there exist a range of explanation approaches, the possibilities for integrating domain knowledge is still lacking, augmenting according interpretable/explainable models. Some research on this has been presented by [96], [97].

Conclusion

In this paper, we reviewed articles that use data-driven models for predictive maintenance and are explicitly motivated by model interpretability. These papers either used intrinsically interpretable models or post-hoc explanations. We argue that there is a number of open research challenges, from which we identified the following as the most crucial ones: Interpretable machine learning requires some form of acknowledged *evaluation measure* in order to be comparable. Furthermore, *easy to use post-hoc explanation methods* are desirable, i.e. explanation methods should not behave like black boxes themselves. In addition, *explanation coverage* and *inclusion of domain knowledge* into explanations should be considered. Finally, we observed that the majority of research on interpretability addresses images. However, time series are omnipresent when monitoring technical systems, e.g. in predictive maintenance. Hence, we see a *strong need for research on interpretable ML/XAI on time series*. In fact, this will be the core topic of our future work.

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